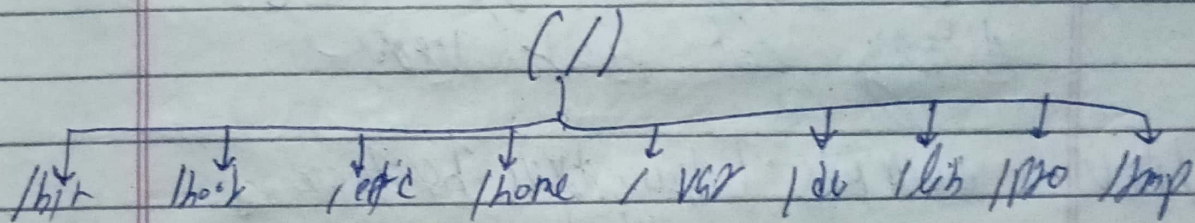


DAY-02 Linux File System Hierarchy & Scenario-Based Practice

PART 1: Linux File System Hierarchy

The Linux file system hierarchy is a standard directory structure that organizes files and data on storage devices, starting from the root directory (/):



* / → The root directory, the top of the hierarchy. Everything starts from this.

* ^{usr} /bin → Essential binary program (like ls, cp)
[mv, cat, etc.]

* /sbin → System administration commands
[used for managing and repairing the system.]

Example:-

/sbin/xboot
/sbin/fsck
/sbin/shutdown

* /boot → This directory we use when your system is startup:-

* /etc - This is for configuration files.

Exmpl:-

/etc/passwd

/etc/hosts

* /home - User home directories -

→ The stores personal files

Exmpl:- /home/surya

* /root - This directory is for Home directory of root user:-

* /dev - device files: [This is the hardware of files]

Exmpl:-

/dev/sda

/dev/hull

* /proc - virtual file system:-

Contain process and kernel information

* you find here primary memory

eg:- /proc/meminfo

all the process related files you are here find!

Exmpl:-

ls -l /proc/meminfo

* /sys
 → It contains kernel and hardware information.

Interface between kernel and device.
 Example:-

/sys
 Contains:
 block
 bus
 class
 dev
 kernel

* /tmp (Temporary)
 This directory contains temporary files.
 Example:

cd /tmp
 touch demo.txt

* /var → variable
 The variable data files such as log files are located in the /var directory.

- /var/log: System log files generated by OS and other applications.
- /var/lib: Contains database and package
- /var/mail: Contains Emails
- /var/tmp: Contains temporary files

* /usr :- The usr directory contains
(user binaries) applications and files used by
user, as opposed to application
and app files used by the
system.

↳ /usr/lib :- shared libraries
↳ /usr/bin :- user binaries [hard disk]
:- Command used by all the
user of the system
located here. [python, gcc]

↳ /sbin :- system binaries :-
Command used by the system
administrator, for system
maintenance purpose.
Example:- ip tables, sshd, passwd,
ifconfig, etc.

* /opt :- optional third-party software
Example:- /opt/google
/opt/jre

* /mnt :- Temporary mount point.
This directory is used to mount a
file system temporarily.

* /media :- Removable media devices

The media directory contains subdirectories
where removable media devices inserted
into the computer are mounted.

* `du -sh /var/log/* 2>/dev/null | sort -h | tail -5`

* `du -sh /var/log/*`

It shows the ~~log~~ all files that can use disk space.

(ii) `2>/dev/null` [Error all hide]

Note:-

0 key keyboard input
1 Normal output
2 Error output

⇒ `/dev/null`
Special device that discards everything
anything sent there disappears -
echo "hello" > /dev/null

(iii) `sort -h`

Pipe operator. Pass out from one command to another.

`sort` :- sort lines

`-h` :- Human-readable sort

10k

50M

25%

without `-h`

→ 10k

258

50M

(iv) `tail` :- display last lines

`-s` :- show last 5 lines.

[`tail -h 5`]

* cat /etc/passwd
where the username used to
identify the system on a whole

* ls -la ~

This command all files and
directory with hidden in your home
directory

$\sim = /home/alan$

$\sim$ = This is shortcut

Part 2: - Scenario-Based Practice

Example Scenario: - Check if a service is running: -

Step 1: - Check service status
system status `systemctl` -
This command is used to check if the service is running or not.

Step 2: - `systemctl list-units --type=services`
List all services whether it is running or not.

Step 3: - Check if service is enabled on the system or is enabled `systemctl`

[Scenario: Service Not Starting]

Step 1: - `systemctl status` `systemd`
This command is used to check if the service is running or not.

Step 2: - `systemctl is-enabled` `systemd`
This command is used to show if the service is enabled on boot. To know if it will start automatically after reboot.

Step 4: - `journalctl -u` `systemd`
`journalctl -u` `systemd` -h 50
This command shows the log of the service and show last 50 lines.

[Scenario 2:- High CPU usage]

Step 1:- Top :-

This command is used to check real time CPU and Mem usage

Step 2: ps aux --sort=-%cpu | head -5
This command is used to show accuracy in the CPU usage high to low (-) but 5 process.

Way point (-) :- use for high to low
(+) :- use for low to high

Step 3:- ps -o pid,ppid,pmem,pcpu,cmd -p [pid]

This command is show a particular process information about p[pid], mem, CPU, how much use.

ps aux --sort=-%cpu # Highest CPU

ps aux --sort=-%cpu # Lowest CPU

ps aux --sort=-%mem # Highest memory

ps aux --sort=-%mem # Low memory

Scenario 3: Finding service logs:-

Step 1:- systemctl status ssh

Check if the service is running or not.

Step 2:- journal -4 ssh -4 50

This command is used to list 50 lines of log ssh.

Step 3: journalctl -4 ssh -f

To see real-time log of ssh:-

[Scenario 4: file permission issue]

Step 1: check

This script - backup.sh is not
execute.

Step 1: ls -l backup.sh

This step command the use to
show the permission of the file

Step 2: - chmod execute permission
at chmod +x backup.sh

This command execute permission
for all the users.

Step 3: verify : ls -l backup.sh

Step 4: Try this it

Command: ./backup.sh